

Highlights

Agentic Lightweight Consensus for Resilient Monitoring and Control of Data-Centre Cyber–Physical Systems

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- A two-path architecture: a deterministic per-tick fuzzy-consensus fast path and a sparse, event-triggered LLM supervisory path.
- A lightweight FPR–OWA consensus operator whose value lies in confidence and anomaly diagnostics, not in dominating MAE.
- Across 1,890 real-data scenarios no aggregator dominates; redundancy and attack type, not the operator, drive accuracy.
- A non-bypassable deterministic verifier reduces executed unsafe actions to zero; without it, all unsafe proposals execute.
- Event-triggered LLM supervision preserves accuracy and SLA safety while cutting deadline misses by $\approx 95\%$ versus intensive use.

Agentic Lightweight Consensus for Resilient Monitoring and Control of Data-Centre Cyber–Physical Systems

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Abstract

Modern data centres operate as tightly coupled cyber–physical systems in which thermal, power, and workload signals must be reconciled in real time to keep monitoring and control loops both responsive and trustworthy. A growing body of work addresses coordination in sensor-rich infrastructures through blockchain-oriented lightweight consensus, yet the agreement problem that dominates data-centre operation is not transaction settlement: it is the derivation of stable, attack-resilient operational state from noisy, heterogeneous, and possibly deceptive local reports. This paper proposes an agentic lightweight consensus architecture that addresses this problem directly. The architecture organises sensing, edge aggregation, zone verification, and facility-level arbitration into a hierarchy of cooperating agents, and grounds its consensus layer in the soft-consensus tradition of fuzzy group decision making: sensor reports are scored for reliability, compared through fuzzy preference relations, ordered by quantifier-guided dominance, and aggregated with ordered weighted averaging under a weak-majority linguistic quantifier. A consensus-reaching control loop monitors agreement, proximity, and temporal consistency, and applies targeted corrective feedback to unreliable reporters. The control architecture is deliberately split into two paths: a deterministic, per-tick fast path (sensing → fuzzy consensus → verifier → safe action or fallback) that owns every safety-critical decision, and a slow, event-triggered supervisory path in which a LangGraph-orchestrated large language model planner reasons over consensus snapshots and proposes occasional supervisory actions that must clear the same non-bypassable verifier. We evaluate the architecture on a real-

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data cyber-physical simulation driven by audited data-centre thermal, workload, and anomaly-timing datasets, across 1,890 scenarios spanning seven deception/anomaly attacks. The results are deliberately nuanced. No single aggregator dominates: simple averaging attains the best global mean absolute error, while FPR-OWA wins under systematic bias and, uniquely, supplies the confidence and anomaly signals on which safe orchestration depends. The verifier proves essential—without it every unsafe proposal is executed, with it none are—and self-refinement increases the number of accepted safe actions at the price of latency. Finally, intensive in-the-loop LLM supervision does not improve estimation accuracy yet inflates loop latency past real-time deadlines, whereas event-triggered supervision preserves estimation quality and zero thermal-SLA violations while cutting the deadline-miss rate by roughly 95%. The contribution is an architecture in which agentic reasoning serves as sparse supervisory intelligence while the safety-critical path stays lightweight, deterministic, fuzzy-consensus driven, and verifier-guarded.

Keywords: Agentic AI, Multi-Agent Systems, Internet of Things, Data Centres, Cyber-Physical Systems, Soft Consensus, Fuzzy Preference Relations, Ordered Weighted Averaging, Trustworthy AI, Large Language Models

1. Introduction

Modern data centres are cyber-physical systems (CPS) in which computational workload, electrical power, thermal conditions, and physical sensing evolve together under tight feedback. Production telemetry studies show that workload, energy, and temperature signals are strongly coupled and vary across spatial and temporal scales that no single monitoring point can capture [1]. Monitoring in this setting is therefore not a passive observability concern but part of the operational control loop: local observations must be interpreted quickly enough to support timely reaction, yet they must also be reconciled across racks, zones, and the whole facility whenever they conflict, drift, or imply facility-level trade-offs. Hierarchical and edge-assisted monitoring platforms have demonstrated that layered collection and local pre-processing are practical in internet data centres [2, 3], and autonomic multi-agent management has long shown that data-centre decisions can be decomposed into interacting local and global decision makers [4, 5]. More recently, learning-based controllers have delivered substantial energy savings in cooling management, while exposing how brittle unconstrained

data-driven actuation can be when thermal safety is at stake [8, 9].

What remains underdeveloped between these layers is the *agreement* layer. The operational question a data-centre CPS must answer at every decision round is which local reports to trust, how to fuse them into a stable zone-level state, and what to do when reports disagree—because of sensor noise, communication faults, or deliberate deception. False-data-injection attacks against sensing channels are a well-studied threat to CPS state estimation and can be crafted to remain stealthy while degrading control quality [10]. Lightweight approximate agreement shows that bounded local disagreement can be achieved on small devices under lossy communication [11], but it offers no principled account of *graded* trust in individual reporters, nor a process view of how agreement should be monitored and repaired over time.

A prominent research line approaches coordination in sensor-rich infrastructures through blockchain, designing lightweight and hierarchical consensus mechanisms that fit constrained devices [12, 17, 13, 14, 15, 19, 20, 18]. This line answers a different question: ledger consensus—Byzantine fault tolerant [16] or authority-based—agrees on the *order and integrity of transactions*, whereas data-centre operation requires agreement on the *current physical state* under noise and deception, at rates and energy budgets where ledger replication is unnecessary. We therefore treat our work not as a competitor but as an alternative coordination paradigm, with blockchain a complementary instrument for audit or settlement (Section 2.5).

Two mature research traditions make such an alternative concrete. The first is the theory of *soft consensus* in fuzzy group decision making (GDM), developed by Kacprzyk, Herrera-Viedma, and collaborators: agreement among experts is treated as a graded, monitored process, measured through consensus and proximity degrees, guided by linguistic quantifiers such as “most” [21, 22, 23, 24, 25, 26, 29], constrained by consistency control [30], and repaired through feedback that targets the participants most responsible for disagreement [31, 32, 33, 34]. This machinery—born for human experts—transfers naturally to sensor agents, whose reports are likewise graded, conflicting, and unequally reliable. The second tradition is *agentic AI*: large language model (LLM) agents that interleave reasoning and acting [44], refine their outputs through verbal self-correction [45], and can be organised into role-specialised multi-agent systems [46]. Agentic orchestration brings flexible, explainable planning to CPS control, but it also imports a new trust problem: an LLM planner can propose unsafe actions. Our position is that LLM autonomy in a CPS must be bounded by a deterministic, non-bypassable verification gate, with the consensus layer supplying the validated evidence

on which both planner and verifier operate.

This paper integrates the two traditions into a single architecture for data-centre CPS monitoring and control, and evaluates it on a real-data cyber-physical simulation. We address four research questions, each answered directly by the campaign of Section 7:

- **RQ1 (estimation).** How do the lightweight fuzzy-consensus operator (FPR-OWA) and standard robust aggregators compare for zone-level thermal state estimation under deception attacks, and what governs estimation quality? We ask which method wins *where*, rather than assuming any single operator dominates.
- **RQ2 (diagnostic value).** Beyond point-estimate error, what distinctive role does FPR-OWA play through the confidence and anomaly signals it produces, and why are these signals—rather than a lower MAE—the reason to retain it in a safety-oriented architecture?
- **RQ3 (safety enforcement).** Does a non-bypassable deterministic verifier eliminate the execution of unsafe LLM-proposed actions, and what does self-refinement add in terms of accepted safe actions versus computational cost?
- **RQ4 (real-time feasibility).** How does the rate of LLM supervision affect estimation accuracy, loop latency, and deadline satisfaction, and can event-triggered supervision retain accuracy and SLA safety while restoring real-time responsiveness?

We deliberately separate three concerns that the literature often conflates: *estimation accuracy* (RQ1–RQ2), *safety enforcement* (RQ3), and *operational latency* (RQ4). The central finding is that they are best served by different mechanisms—robust fuzzy consensus for the first, a deterministic verifier for the second, and sparse event-triggered LLM supervision for the third—rather than by placing a single powerful reasoner in the loop.

The contributions are fourfold:

1. **C1: Architecture.** A blockchain-independent, hierarchical agentic architecture for data-centre CPS sensing and control, in which sensing agents, edge consensus agents, zone verifiers, an LLM planner, and a deterministic safety verifier cooperate under explicit trust-oriented design principles (Section 3).

2. **C2: Consensus methodology.** A lightweight consensus operator (FPR–OWA) that scores reporter reliability, builds a fuzzy preference relation over reporters, orders them by quantifier-guided dominance, and aggregates with OWA weights induced by a weak-majority linguistic quantifier, together with a zone-level consensus-reaching loop with proximity, temporal consistency, and adaptive feedback rooted in the Herrera-Viedma soft-consensus tradition (Section 4).
3. **C3: Verifier-gated agentic control.** A control methodology in which every LLM-proposed action must pass a deterministic verifier enforcing freshness, confidence, actuator, and thermal-safety constraints, with Reflexion-style refinement cycles and complete execution traceability (Section 5).
4. **C4: Empirical evidence.** A real-data evaluation over 1,890 scenarios and two real-time supervision regimes that disentangles estimation accuracy, safety enforcement, and latency, and that supports a measured rather than promotional account of when fuzzy consensus, the verifier, and LLM supervision each help (Sections 6–7).

The remainder of the paper is organised as follows. Section 2 reviews related work and states the research gap. Section 3 presents the architecture, the system model, and the fault model. Section 4 formalises the consensus methodology, and Section 5 the verifier-gated agentic control loop. Section 6 describes the experimental setup, Section 7 reports the results, and Section 8 discusses implications and threats to validity. Section 10 concludes.

2. Related Work

Four research lines converge on this work: consensus for blockchain–IoT integration, hierarchical monitoring and multi-agent management of data centres, soft consensus in fuzzy group decision making, and LLM-based agentic systems. They have largely progressed independently; we review each and then state the gap.

2.1. Consensus in Blockchain–IoT Integration

Blockchain–IoT integration has produced a rich catalogue of consensus mechanisms for constrained devices: authority-based validation, edge-hosted ledgers, hybrid on-chain/off-chain patterns, and energy-recycling schemes [13, 14, 15, 19, 20, 18, 17]. Closest to our starting point, Ramachandran’s three-tier hierarchical framework matches device classes to consensus roles, with district PBFT [16] under city-wide proof of authority [12]. All share one

object of agreement—the ledger—and guarantee transaction ordering, immutability, and quorum bounds; the problem here, estimating physical state from graded, partially deceptive reports at every control round, lies outside that object, which is why we treat ledger consensus as complement rather than baseline (Section 2.5). In the non-ledger space, Zhang et al. design approximate consensus for small devices in lossy networks [11]; we share that lightweight spirit but add the treatment of reporter trust it lacks.

2.2. Data-Centre Monitoring, Multi-Agent Management, and Safe Control

Hierarchical sensing is established practice in data centres. Jiang et al. partition an internet data centre into logical grids whose edge nodes pre-process environmental and host telemetry before escalation [2]; Yan et al. combine edge computing, wireless sensing, and IPMI-derived measurements for environmental monitoring [3]. On the management side, the Unity system decomposes power and performance management into interacting agents [4], within the broader autonomic-computing programme of decentralised, self-managing elements [5]. Multi-layer IoT/edge coordination through compact summaries rather than raw-state collection is likewise well supported [6, 7]. Production-trace characterisation confirms the coupled dynamics of workload, energy, and temperature that such hierarchies must track [1].

On the actuation side, deep reinforcement learning has reduced cooling energy substantially, but unconstrained exploration violates thermal safety; physics-guided safe RL mitigates this by rectifying recommended actions against a learned thermal model [8], and event-driven formulations reduce unnecessary control operations [9]. These results motivate two design commitments of our architecture: control proposals must be validated against explicit safety constraints before actuation, and the state that feeds control must itself be validated against noise and deception [10].

2.3. Soft Consensus in Fuzzy Group Decision Making

The consensus layer of this paper draws on a four-decade research programme on consensus reaching in fuzzy GDM. Its foundations are the calculus of linguistically quantified propositions [21], the OWA operator as the canonical implementation of relative quantifiers such as “most” [22], and the notion of a *soft* degree of consensus—most of the important individuals agreeing on most of the relevant options—introduced by Kacprzyk and colleagues [23, 24, 25]. Herrera et al. brought this view to linguistic assessments [26], and Bordogna et al. modelled linguistic consensus through

OWA aggregation [27]. Fuzzy preference relations became the standard representation of graded pairwise judgements, with integration results across preference formats [28] and consensus models that operate across heterogeneous structures [29].

Three later developments matter most here. First, consensus models were equipped with *guidance*: consensus degrees and proximity measures computed at several representation levels drive automatic feedback that tells specific participants how to revise specific assessments [29, 31]. Second, *consistency control* was combined with consensus so that agreement is not reached at the price of self-contradictory preferences, including under incomplete information [30, 33]. Third, the field produced systematic analyses and reviews—of soft consensus models in fuzzy environments [34, 32, 35], of consensus measures [36], of minimum-adjustment feedback mechanisms [37], and of dynamic consensus and opinion dynamics in networked settings [38]—together with operator-level tools such as type-1 OWA for multi-granular linguistic contexts [39] and methodologies for unbalanced linguistic term sets [40]. Applications include collaborative knowledge engineering under linguistic fuzzy consensus [41].

This tradition treats consensus as a monitored *process* over unequally reliable participants—exactly the structure of a zone of sensor agents under deception. Its transfer to machine-to-machine settings, however, remains largely unexplored: consensus-reaching models assume human experts revising opinions, whereas sensor agents require automated analogues of reliability, preference, dominance, and feedback. Constructing those analogues is the methodological core of this paper.

2.4. LLM-Based Agents and Trustworthy Agentic Orchestration

LLM-based agents couple language-model reasoning with tool use and environment interaction [46]. ReAct interleaves explicit reasoning traces with actions, improving groundedness and auditability [44]; Reflexion adds verbal reinforcement, letting agents revise behaviour from feedback without weight updates [45]. Reasoning-quality mechanisms such as chain-of-thought prompting [42] and self-consistency over sampled reasoning paths [43] establish the sampling-and-agreement intuition that softer, fuzzy notions of consensus generalise. In safety-critical domains, the emerging consensus is that reasoning traces alone do not establish trustworthiness; explicit verification stages and human-auditable evidence are required. We adopt this position architecturally: the LLM planner is a proposer, never an executor, and a deterministic verifier owns the actuation decision.

Table 1: Agentic lightweight consensus vs. blockchain-IoT ledger consensus: different objects of agreement, hence complementary.

Dimension	Blockchain-IoT consensus (e.g., [12, 17])	Agentic lightweight consensus (this work)
Object of agreement	Order and integrity of transactions in a replicated ledger	Current operational state of a monitored zone
Trust target	Validator honesty under Byzantine quorum bounds ($f < n/3$) [16]	Graded reliability of individual sensor reports under noise and deception
Primary guarantee	Immutability, auditability, finality of committed records	Stability and attack-resilience of state estimates feeding control
Adversary model	Byzantine validators, Sybil identities, long-range attacks	Faulty or compromised sensors (bias, drift, freeze, replay, scaling)
Coordination cost	Multi-round message exchange, ledger storage, anchoring	Local computation on zone-sized report vectors; no replicated state
Latency profile	Block/anchoring intervals (sub-second to minutes)	Per-round aggregation at telemetry rate
Role of human oversight	Governance of validator sets and policies	Auditable traces, escalation, verifier-policy configuration
Complementarity	Can notarise validated state summaries and audits	Can supply validated state worth notarising

2.5. Research Gap and Positioning

The literature thus supplies hierarchical sensing [2, 3], multi-agent data-centre management [4, 5], lightweight approximate agreement [11], a mature theory of monitored, feedback-driven consensus [34], and agentic orchestration patterns [44, 45]. No existing work, to the best of our knowledge, integrates these ingredients into a single blockchain-independent architecture in which (i) the consensus object is operational state rather than a ledger, (ii) reporter trust is modelled through fuzzy preference relations and quantifier-guided dominance, (iii) consensus is monitored and repaired as a process, and (iv) LLM-driven control is gated by deterministic verification. Table 1 summarises the qualitative positioning of this proposal with respect to blockchain-IoT consensus; the comparison is conceptual by design, since the two paradigms answer different agreement problems.

3. Architecture and System Model

This section presents the proposed architecture, the agents it comprises, and the system and fault models under which it operates.

3.1. Design Principles

Five principles drive the design.

- P1 — Consensus on state, not on transactions.** The agreement mechanism validates operational state at every decision round; it maintains no ledger and requires no global ordering of events.
- P2 — Lightweight, quantifier-guided aggregation.** Agreement is computed locally on zone-sized groups of reports with closed-form operators (medians, fuzzy preference relations, OWA weights), so that the consensus step remains compatible with edge-class hardware.
- P3 — Consensus as a monitored process.** Following the soft-consensus tradition [34], agreement is measured (consensus degree, proximity, consistency) and repaired (targeted feedback) rather than assumed after a single aggregation step.
- P4 — Bounded agentic autonomy.** LLM-based planning is a proposal mechanism. A deterministic verifier owns the decision to actuate; no proposed action can bypass it.
- P5 — Traceability by construction.** Every consensus snapshot, planner proposal, verifier decision, and refinement cycle is logged in structured, replayable execution traces.

3.2. Layered Architecture and Agent Roles

Figure 1 depicts the architecture as two interacting paths. The *fast path* is deterministic and runs at every control tick: sensing agents feed the edge fuzzy-consensus operator, whose validated snapshot is checked by a non-bypassable verifier that releases either a safe control action or a conservative deterministic fallback. The *slow path* is an event-triggered supervisory loop in which a LangGraph-orchestrated LLM planner reasons over consensus snapshots and proposes occasional supervisory actions; crucially, these proposals re-enter the *same* verifier before they can affect actuation. A trustworthiness-and-traceability plane crosses both paths.

Physical and sensing layer. Sensor agents attached to racks, hosts, and power-distribution interfaces emit per-round observations of thermal, power, and workload variables, in line with hierarchical data-centre monitoring practice [2, 3]. Actuator interfaces (fan speed, cooling setpoints) close the physical loop.

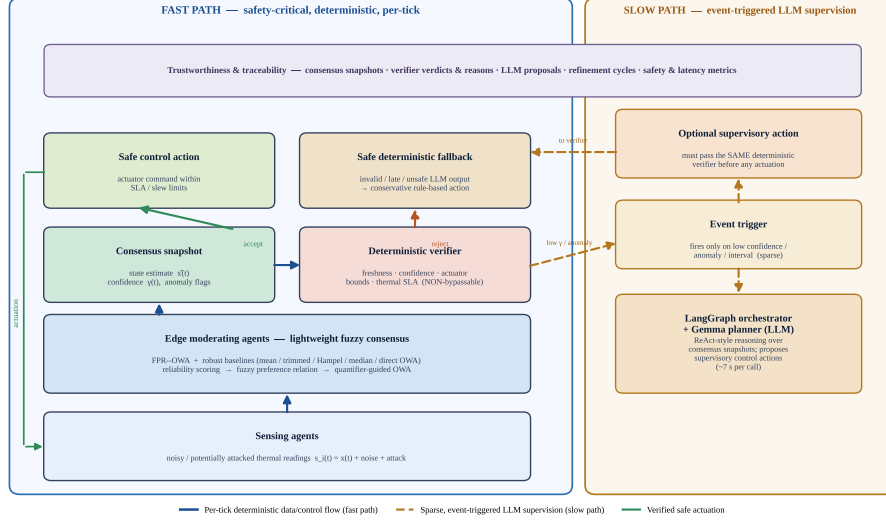


Figure 1: Two-path architecture. The deterministic **fast path** (left) runs every tick: sensing $\rightarrow$ fuzzy consensus $\rightarrow$ verifier $\rightarrow$ safe action or fallback. The sparse **slow path** (right) is an event-triggered LangGraph/Gemma planner whose proposals must clear the same non-bypassable verifier. A traceability plane records all decisions.

Edge consensus layer. Edge moderating agents collect the reports of one zone and execute the FPR-OWA lightweight consensus operator (Section 4.3), producing a zone state estimate together with diagnostic metadata: per-sensor reliability scores, anomaly flags, and an aggregate confidence value. The result is packaged as a *consensus snapshot*, the only sensing artefact visible to the upper layers.

Zone verification and coordination layer. Zone verifier agents monitor the consensus process over time: they compute consensus degree, per-agent proximity, and temporal consistency, and trigger the adaptive feedback actions of Section 4.5 (re-sampling, down-weighting, delayed inclusion, persistent-anomaly flagging) when agreement degrades. This layer is the sensor-agent analogue of the moderator in consensus-reaching models [31, 30].

Agentic orchestration and control layer (slow path). A LangGraph-orchestrated LLM planner agent reasons over consensus snapshots under a ReAct-style scheme [44] and proposes structured *supervisory* control actions. This layer is invoked sparsely and asynchronously—triggered by low confidence, anomaly flags, or a minimum inter-invocation interval—rather than on every tick, because (as Section 7 shows) a single LLM call costs seconds and frequent

invocation degrades real-time responsiveness without improving estimation. Every proposal, whenever it is generated, is checked by the same deterministic verifier of the fast path, which accepts it, rejects it, or returns machine-readable feedback that drives optional Reflexion-style self-refinement [45] for at most K cycles (Section 5).

Facility operation layer. A facility arbiter resolves cross-zone trade-offs (thermal risk versus energy cost, workload placement), consistent with the local-global decomposition of autonomic data-centre management [4, 5], and exposes dashboards for human oversight and escalation.

Trustworthiness and traceability plane. Crossing all layers, a traceability service records consensus snapshots, reliability vectors, planner prompts and outputs, verifier decisions with reasons, refinement cycles, and safety counters, yielding an auditable, replayable account of every control decision (P5).

3.3. System Model

Let $\mathcal{Z}$ be the set of zones. Zone $z \in \mathcal{Z}$ contains n_z sensing agents indexed by $E_z = \{1, \dots, n_z\}$. Time is discretised into decision rounds $t = 1, 2, \dots$ aligned with the telemetry period. At round t , agent $i \in E_z$ reports a measurement

$$s_i(t) = x_z(t) + \varepsilon_i(t) + a_i(t), \quad (1)$$

where $x_z(t)$ is the latent zone state (in the reference use case, a zone thermal variable), $\varepsilon_i(t)$ is zero-mean sensing noise, and $a_i(t)$ is an attack term, zero for honest reports. The exposition uses a scalar state for clarity; the operators extend componentwise to state vectors in $\mathbb{R}^d$ combining thermal, power, and workload features.

Communication between sensing and edge agents is local, frequent, and loss-prone; messages may be delayed, dropped, or duplicated, in the spirit of lightweight agreement under lossy networks [11]. Zone-to-facility communication is less frequent and carries only consensus snapshots and verified control intents. No global clock or reliable transport is assumed; each snapshot carries a timestamp, and staleness is policed by the verifier (Section 5.2).

3.4. Fault and Attack Model

We consider a deception adversary who compromises a fraction $\beta_a \in [0, 1)$ of the sensing agents of a zone and manipulates their reports through the attack term $a_i(t)$ in (1). Table 2 lists the attack taxonomy, chosen to cover the canonical false-data-injection patterns studied in CPS security [10] as they manifest in slow-dynamics thermal telemetry. Benign faults (missing

Table 2: Deception-attack taxonomy. A fraction β_a of a zone’s agents is compromised, possibly intermittently.

Attack	Form of $a_i(t)$	Operational effect
Bias	constant offset b	Systematic over/under-reporting that shifts the zone estimate
Drift	ramp $\delta(t - t_0)$	Slow poisoning that mimics gradual thermal change and evades thresholding
Freeze	$s_i(t) \equiv s_i(t_0)$	Stale value masking real dynamics (replay of own last reading)
Replay	$s_i(t) = s_i(t - \Delta)$	Re-injection of a past window, plausible in distribution but desynchronised
Scaling	multiplicative $(\gamma - 1)x_z(t)$	Proportional distortion preserving trends but corrupting magnitude
Mixed	time-varying combination	Alternating patterns that defeat single-statistic defences

and delayed reports, transient noise bursts) are handled by the same machinery, since the consensus layer makes no attempt to distinguish malice from malfunction: both surface as unreliable reports.

The defender’s goal is not exact agreement under arbitrary Byzantine behaviour—a guarantee that ledger consensus buys at a cost incompatible with per-round telemetry—but graceful degradation: producing a zone estimate whose error grows slowly with β_a and attack magnitude, while flagging the implicated reporters for the feedback loop of Section 4.5.

4. Lightweight Consensus Methodology

This section formalises the consensus layer: the per-round FPR-OWA operator executed by edge agents (Section 4.3) and the zone-level consensus-reaching loop that monitors and repairs agreement over time (Section 4.5). The design transfers, to sensor agents, four ingredients of soft-consensus GDM: graded reliability in place of expert importance, fuzzy preference relations in place of pairwise expert judgements, quantifier-guided dominance in place of selection degrees, and feedback in place of moderator advice.

4.1. Reliability Scoring

Let $\mathbf{s}(t) = (s_1(t), \dots, s_{n_z}(t))$ be the report vector of zone z at round t (zone subscripts are omitted where unambiguous). A robust centre and scale are computed first:

$$m(t) = \text{median}_i s_i(t), \quad \hat{\sigma}(t) = 1.4826 \cdot \text{MAD}_i s_i(t) + \epsilon, \quad (2)$$

where MAD is the median absolute deviation, the constant 1.4826 makes $\hat{\sigma}$ consistent with the standard deviation under Gaussian noise, and $\epsilon > 0$ prevents degeneracy when reports coincide. Each agent then receives a reliability score

$$\rho_i(t) = \exp\left(-\frac{|s_i(t) - m(t)|}{\hat{\sigma}(t)}\right) \in (0, 1]. \quad (3)$$

Reliability decays exponentially with robust-standardised distance from the group centre: agents near the consensus of their peers score close to one, while outliers—whether noisy or attacked—are discounted smoothly rather than excluded by a hard threshold. This is the sensor-agent analogue of the graded importance weights of soft consensus [23, 24].

4.2. Fuzzy Preference Relation over Reporters

Pointwise reliability ignores the relational structure of trust: what matters for ordering reporters is how much more credible one is than another. Following the representation of graded pairwise judgements by fuzzy preference relations [28, 29], we build $P(t) = [p_{ij}(t)] \in [0, 1]^{n_z \times n_z}$ with

$$p_{ij}(t) = \frac{1}{1 + \exp(-\kappa(\rho_i(t) - \rho_j(t)))}, \quad p_{ii}(t) = 0.5, \quad (4)$$

where $\kappa > 0$ controls the sharpness of the comparison. The relation is additively reciprocal, $p_{ij} + p_{ji} = 1$, and $p_{ij} > 0.5$ reads “report i is more credible than report j ”. The logistic link makes the preference robust to small reliability differences (which map to values near 0.5) while saturating for clear-cut cases.

From $P(t)$, each agent receives a dominance degree

$$\delta_i(t) = \frac{1}{n_z - 1} \sum_{j \neq i} p_{ij}(t) \in [0, 1], \quad (5)$$

the average strength with which i is preferred to its peers. This is the operational counterpart of the quantifier-guided dominance degrees used to exploit fuzzy preference relations in GDM selection processes [28, 29]; it induces the credibility ordering used by the aggregation step.

4.3. Quantifier-Guided OWA Aggregation

Let σ be the permutation of E_z that sorts dominance degrees in non-increasing order, $\delta_{\sigma(1)} \geq \dots \geq \delta_{\sigma(n_z)}$. Aggregation weights are derived from

a regular non-decreasing linguistic quantifier in the parametric family of Zadeh [21],

$$Q_{a,b}(r) = \begin{cases} 0, & r < a, \\ \frac{r-a}{b-a}, & a \leq r \leq b, \\ 1, & r > b, \end{cases} \quad 0 \leq a < b \leq 1, \quad (6)$$

instantiated with $(a, b) = (0.3, 0.8)$, the classical parametrisation of the weak-majority quantifier “most” [26, 29]. The OWA weights of Yager [22],

$$w_k = Q_{a,b}\left(\frac{k}{n_z}\right) - Q_{a,b}\left(\frac{k-1}{n_z}\right), \quad k = 1, \dots, n_z, \quad (7)$$

are assigned along the credibility ordering so that the mass prescribed by the quantifier falls on the most dominant reports, and the zone estimate is

$$\hat{x}_z(t) = \sum_{k=1}^{n_z} w_k s_{\sigma(k)}(t). \quad (8)$$

The semantics is a fuzzy-majority fusion: the validated state is the value supported by *most of the credible* reporters, rather than the unweighted average (vulnerable to coordinated bias) or the bare median (which discards the graded structure of trust). Because ordering is dominance-induced rather than value-induced, the operator behaves as an induced OWA whose order-inducing variable is relational credibility—the key difference from a direct OWA on sorted measurement values, which we retain as an ablation baseline.

4.4. Diagnostics and Consensus Snapshot

Alongside the estimate, the edge agent emits diagnostics. Agents whose reliability falls below an adaptive threshold are flagged as anomalous:

$$\begin{aligned} A_z(t) &= \{ i \in E_z : \rho_i(t) < \tau_\rho(t) \}, \\ \tau_\rho(t) &= \max(\underline{\tau}, \text{median}_i \rho_i(t) - 2 \text{std}_i \rho_i(t)), \end{aligned} \quad (9)$$

with a floor $\underline{\tau}$ that keeps the threshold meaningful when reliabilities are uniformly high. The zone confidence is the mean reliability $\gamma_z(t) = \frac{1}{n_z} \sum_i \rho_i(t)$. The *consensus snapshot*

$$\chi_z(t) = (\hat{x}_z(t), \gamma_z(t), A_z(t), \boldsymbol{\rho}(t), P(t), t) \quad (10)$$

is the typed artefact passed upward; it carries enough evidence for the verifier to police staleness and confidence, and for the traceability plane to

support post-hoc audit. The operator costs $O(n_z^2)$ arithmetic per zone and round, dominated by the preference relation; for zone-sized groups (tens of sensors) this is negligible on edge-class hardware and involves no inter-zone communication, in contrast with the $O(n^2)$ *message* complexity of PBFT-style quorums [16].

4.5. Zone-Level Consensus Reaching and Adaptive Feedback

A single aggregation round, however robust, cannot distinguish an agent that is briefly noisy from one that is persistently deceptive, nor can it recover information from agents that are systematically down-weighted. The zone verifier therefore runs a consensus-reaching process (CRP) over rounds, adapting the guidance-and-feedback scheme of soft-consensus GDM [29, 30, 31, 33] to sensor agents.

Consensus degree and proximity. Pairwise similarity between reports is measured by soft coincidence,

$$s_{ij}(t) = \max\left(0, 1 - \frac{|s_i(t) - s_j(t)|}{M_z}\right), \quad (11)$$

with $M_z > 0$ a zone normalisation constant (e.g., the admissible operating range of the monitored variable). Agent proximity and the zone consensus degree follow the three-level construction of consensus models [31]:

$$p_i(t) = \frac{1}{n_z - 1} \sum_{j \neq i} s_{ij}(t), \quad C_z(t) = \frac{1}{n_z} \sum_{i \in E_z} p_i(t). \quad (12)$$

Temporal consistency. Mirroring the consistency control that prevents agreement from masking self-contradiction [30, 33], each agent is also scored for coherence with its own recent behaviour. Let $\tilde{s}_i(t)$ be a short-horizon prediction of agent i 's report (in the simplest instantiation, an exponentially weighted moving average of its past reports). The temporal consistency score is

$$k_i(t) = \max\left(0, 1 - \frac{|s_i(t) - \tilde{s}_i(t)|}{M_z}\right), \quad (13)$$

and the combined per-agent reliability used by the CRP blends social agreement and self-coherence,

$$r_i(t) = \alpha p_i(t) + (1 - \alpha) k_i(t), \quad \alpha \in [0, 1]. \quad (14)$$

Replay and freeze attacks are the motivating cases: a frozen sensor can remain socially plausible for a while yet fails consistency once the zone dynamics move, whereas a noisy-but-honest sensor fails proximity transiently while staying self-consistent.

Acceptance and feedback. Given thresholds τ_c (zone consensus) and τ_r (agent reliability), the verifier proceeds as follows at each round:

1. compute $\chi_z(t)$ via Sections 4.1–4.4, and $C_z(t)$, $r_i(t)$ via (12)–(14);
2. if $C_z(t) \geq \tau_c$, accept and forward $\chi_z(t)$;
3. otherwise, build the feedback set $F_z(t) = \{i : r_i(t) < \tau_r\}$ and apply to each $i \in F_z(t)$ one or more corrective actions: *re-sampling* (request a fresh report), *temporary down-weighting* (attenuate i ’s influence in the next aggregation), *delayed inclusion* (exclude i for one round), or a *persistent-anomaly flag* when r_i remains below τ_r for L consecutive rounds, which escalates the agent to the facility layer for inspection;
4. re-validate and forward the resulting snapshot, recording all feedback actions in the trace.

This is the sensor-agent analogue of moderator advice generation: the process targets the participants most responsible for disagreement instead of recomputing blindly over the whole zone [31, 33]. The CRP costs the same $O(n_z^2)$ per round as the operator itself and never blocks the control loop: when consensus cannot be restored within a round, the snapshot is forwarded with degraded confidence, and the confidence gate of the verifier (Section 5.2) decides whether control may proceed.

5. Verifier-Gated Agentic Control

The control side of the architecture turns validated state into safe actuation. It deliberately separates a flexible, explainable *proposer* from a rigid, deterministic *gatekeeper*.

5.1. Event-Triggered ReAct Planner Agent

The planner is an LLM agent orchestrated with LangGraph and backed, in our implementation, by a Gemma instruction-tuned model served behind a standard inference endpoint. Operating under a ReAct-style scheme [44], it receives the consensus snapshot $\chi_z(t)$ (and, where relevant, recent history and the current actuator state), reasons explicitly about the thermal and energy situation of the zone, and emits a *structured* supervisory action

$$u_z(t) = (\Delta f, \theta_{\text{set}}, \text{rationale}, \text{used_consensus}), \quad (15)$$

where Δf is a fan-speed adjustment, θ_{set} a cooling setpoint, *rationale* a natural-language justification retained for audit, and *used_consensus* a declaration that the proposal was grounded in the validated snapshot rather

than in raw or remembered values. Output is constrained to a typed JSON schema; malformed or schema-violating generations do not stall the loop—they trigger the deterministic fallback (below) and are recorded as handled events, not crashes.

Crucially, the planner is *not* placed on the per-tick critical path. It is invoked by an event-triggered policy parameterised by a confidence threshold, an anomaly-count threshold, a temperature margin, and a minimum inter-invocation interval; between triggers the fast path runs alone. The planner is also model-agnostic: any instruction-following LLM can fill the role, and—by construction—the architecture’s *safety* properties depend on the verifier, not on the planner behaving well.

5.2. Deterministic Verifier Agent

The verifier V is a deterministic rule-based agent—no learned components—that evaluates every proposed action against a declarative policy. Its checks fall into two groups.

Process checks establish that the proposal is procedurally trustworthy: the planner declared grounding in the consensus snapshot; the snapshot is fresh, $t_{\text{now}} - t \leq \tau_{\text{age}}$; and the zone confidence clears the floor, $\gamma_z(t) \geq \tau_\gamma$.

Safety checks establish that the action is physically admissible: actuator values within configured bounds; slew rates within limits; and the predicted post-action zone temperature—obtained from a conservative first-order thermal model—compliant with the service-level threshold T_{SLA} , with a stricter emergency rule activating at T_{emerg} .

The verdict is

$$V(u_z(t), \chi_z(t), \text{actuators}) \in \{\text{ACCEPT}, \text{REJECT}\} \quad \text{with reasons } R(t), \quad (16)$$

where $R(t)$ is a machine-readable list of violated checks. Only ACCEPTed actions reach the actuator interfaces.

Non-bypassability. The actuation path is wired exclusively through V : planner outputs have no side effects, and the actuator interface consumes only verifier-signed actions. Liveness is preserved by a deterministic fallback policy (a conservative rule-based controller) that takes over when no proposal is accepted within the refinement budget, so safety never depends on the LLM behaving well and control never stalls on the LLM behaving badly.

5.3. Self-Refinement Cycles

On REJECT, the reasons $R(t)$ are fed back to the planner, which may revise its proposal—a Reflexion-style verbal-feedback loop [45]—for at most

K cycles:

$$u_z^{(c+1)}(t) = \pi_{\text{LLM}}(\chi_z(t), R^{(c)}(t)), \quad c = 0, \dots, K - 1. \quad (17)$$

Refinement converts the verifier from a pure filter into a teacher: the planner learns, within the episode, which constraint it violated and how to satisfy it, without any weight update. The evaluation (Section 6) treats the planner variants—deterministic policy only, ReAct without verifier, ReAct with verifier, ReAct with verifier and self-refinement—as ablation conditions for RQ3.

5.4. Trustworthiness and Traceability

Every round appends to the execution trace: the consensus snapshot and its diagnostics, CRP feedback actions, planner prompt and raw output, verifier verdict and reasons, refinement cycles, the executed action, and timing and token accounting. From these traces the framework derives the safety indicators used in the evaluation—rates of unsafe actions *proposed*, *blocked*, and (the critical one) *executed*, acceptance rates, and refinement-cycle counts—as well as latency and overhead profiles. The trace design follows the same auditability rationale as structured reasoning traces [42] and agreement-based stability signals over sampled outputs [43], but anchors trust in recorded *decisions* rather than in the persuasiveness of generated explanations.

6. Experimental Setup

The architecture is evaluated on a real-data cyber–physical simulation: a hierarchical IoT/MAS data-centre testbed in which sensing agents, edge moderating agents, and the verifier-gated orchestrator operate over telemetry drawn from audited public datasets. The study is a simulation, not a physical deployment; this scoping is intentional and its consequences are discussed in Section 9.

6.1. Data and Their Audited Roles

Dataset names were treated as hypotheses and validated by an automated audit before use; each dataset is admitted only for the role its schema supports (Table 3). Thermal calibration—the hot-corridor temperature signal that the consensus layer estimates—comes from the Kaggle *Data Centre Warm/Hot Corridor Temperature Prediction* dataset (fields `TLHC`, `T_MEAS-*`, `P_cu-*`). The ENACT edge–cloud continuum telemetry (Zenodo record

Table 3: Datasets and their audited roles. Each is used only for the role its schema supports; the CubeSat cybersecurity telemetry contributes anomaly *timing* only.

Dataset	Source	Role	Temp.	Work./Pow.	Attack timing	License
Data Centre Warm/Hot Corridor Temperature Prediction	Kaggle	Thermal calibration (TLHC, T_MEAS-*, P_cu-*)	✓	—	—	Kaggle dataset terms
ENACT Edge-Cloud Continuum Telemetry	Zenodo rec. 18920397	Workload / power context (CPU%, MEM, Energy W)	—	✓	—	Record-specific
CuCD-ID (CubeSat Cybersecurity, Intrusion Detection)	Mendeley Data	Anomaly / deception <i>timing</i> only	—	—	✓	CC BY 4.0

18920397; CPU%, MEM, Energy W) is consumed as a power/workload proxy in the simulation (240 rows per scenario, recorded in each trace’s metadata); the audit marks it usable for power but *not* for temperature, so it is never used as a thermal signal. Because the reported error metrics evaluate the thermal estimate, ENACT enters as simulation context rather than as an evaluated target. Anomaly and deception *timing* is taken from CuCD-ID, a CubeSat cybersecurity intrusion-detection dataset (Mendeley Data, CC BY 4.0); per the audit’s mapping rule, this cyber dataset contributes only the temporal placement of anomaly/attack episodes and is never interpreted as a thermal measurement. Google Cluster Data, Building Data Genome Project 2, and CICIOT2023 were considered as candidates/substitutes but were not the datasets used in the final run. The Kaggle thermal values appear standardised; all error metrics are therefore reported in dataset units unless inverse-scaling metadata is available.

6.2. Scenario Design

The full campaign comprises 1,890 scenarios producing 11,340 per-method metric rows, 30,240 agentic-safety rows, and 41,580 latency rows. Scenarios vary seven deception/anomaly attack types (Table 2: bias, drift, freeze, mixed, replay, scaling, spike), the attacked-sensor fraction $\beta_a \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$, sensor noise levels $\{0.1, 0.25, 0.5\}$, zone sizes $n_z \in \{5, 9, 13\}$, OWA parameters $(\alpha, \beta) \in \{(0.2, 0.7), (0.3, 0.8)\}$, and three random seeds $\{11, 42, 101\}$. The thermal response follows a first-order model

with disturbance, and attacks are activated over a bounded fraction of each episode rather than for its entire duration.

6.3. Consensus Methods and Agentic Variants

Six aggregators are compared under identical conditions: arithmetic mean, trimmed mean, Hampel-filter-then-mean, median, direct OWA (value-ordered, same quantifier), and the proposed *FPR-OWA* (dominance-ordered, with reliability diagnostics). Four agentic-control variants instantiate the safety ablation: *deterministic policy only*, *ReAct without verifier*, *ReAct with verifier*, and *ReAct with verifier and self-refinement*. Two real-time supervision regimes are studied separately (Section 7.4): an *intensive* regime that invokes the LLM aggressively, and an *event-triggered* regime with a restrictive policy (confidence threshold 0.35, anomaly threshold 3, temperature margin 0.02, minimum inter-invocation interval 20 ticks, no real-time refinement), each evaluated against deadline budgets of {100, 250, 500, 1000, 5000} ms.

6.4. Metrics

Estimation quality is measured by mean absolute error (MAE), root mean square error (RMSE), median absolute error, the 95th-percentile absolute error (P95 AE), and maximum absolute error, all in dataset units, plus the mean number of anomaly flags and the mean confidence emitted per method. Safety is measured by the counts of unsafe actions *proposed*, *blocked* by the verifier, and—decisively—*executed*, together with accepted safe actions and LLM/verifier call counts. Latency is reported per decision for the consensus operators and per planner+verifier invocation for the agentic variants; the real-time study additionally reports loop-latency percentiles, LLM invocation and failure rates, deadline-miss rates, and thermal-SLA violation rates.

6.5. Implementation

The fast path is implemented in Python with vectorised NumPy operators. The orchestrator is built on LangGraph with a Gemma planner served behind an OpenAI-compatible endpoint; the verifier is a dependency-free rule engine driven by a declarative SLA policy (T_{SLA} , T_{emerg} , τ_{age} , τ_{γ} , actuator bounds and slew limits, refinement budget K). All runs are seeded and every scenario emits a full execution trace, supporting exact replay and the post-hoc derivation of all reported metrics from measured files only.

Table 4: Global consensus performance over all 1,890 scenarios (dataset units; lower is better; best per column in bold). Only FPR-OWA emits anomaly flags and a non-degenerate confidence.

Method	MAE	RMSE	Median AE	P95 AE	Max AE	Anomaly flags	Confidence
Mean	1.5366	1.6210	1.4226	2.4419	2.8048	–	1.000
Trimmed mean	1.5382	1.6216	1.4255	2.4370	2.7936	–	1.000
Hampel+mean	1.5464	1.6247	1.4330	2.4044	2.8681	–	1.000
Direct OWA	1.5668	1.6431	1.4619	2.3906	2.7126	–	1.000
Median	1.5925	1.6703	1.5063	2.3952	2.6942	–	1.000
FPR-OWA	1.6171	1.7071	1.5437	2.4847	2.8345	1.109	0.532

7. Results

We report measured results only. Throughout, “best” denotes the minimum-error method for the stated slice; no aggregate is projected or smoothed.

7.1. RQ1: How the Consensus Operators Compare

Averaged over all 1,890 scenarios, the six aggregators are close and *no method dominates* (Table 4). Plain *mean* attains the best global MAE (1.5366) and RMSE (1.6210), with trimmed mean and Hampel-mean within 0.01; FPR-OWA has the highest (1.6171). We report this plainly: for unstructured global error, simple robust averaging is hard to beat on this data. The tail, however, reorders the ranking—on P95 AE and maximum absolute error, *direct OWA* (2.3906) and *median* (2.3952) lead the mean (2.4419)—so quantifier-guided aggregation helps where it is designed to, in the worst case, even when it trails on the average.

Method advantage is also attack-dependent (Fig. 2): FPR-OWA is the single best aggregator under *bias* (1.6468 vs. mean 1.7763), where a systematic offset sharpens the reliability ranking; Hampel-mean leads under *spike* (1.4716) and *mixed*; mean leads under drift, freeze, replay, and scaling; and at the heaviest compromise ($\beta_a = 0.4$) the *median* becomes most competitive (1.6133). Yet the dominant driver of accuracy is not the operator but sensor redundancy (Table 5): moving from 5 to 13 sensors per zone cuts mean MAE from 1.8268 to 1.1522 (37%), far more than any between-method gap. The answer to RQ1 is therefore that estimation quality is governed first by redundancy and attack regime, and only secondarily by the aggregator; the fast path should match the aggregator to the threat rather than fix a single “winner”.

7.2. RQ2: The Diagnostic Value of FPR-OWA

If FPR-OWA does not minimise MAE, why retain it? Because it is the only method in the comparison that produces a usable trust signal. The

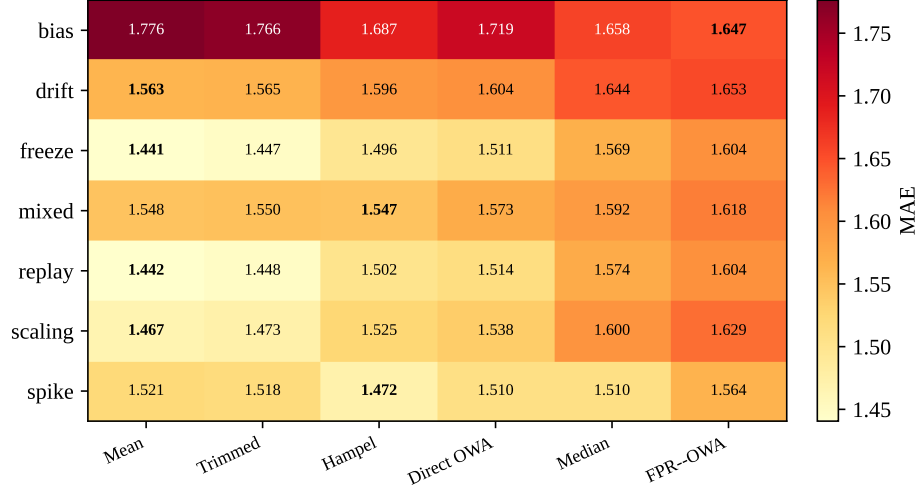


Figure 2: MAE by attack type and method (bold = best per attack). Advantage is attack-dependent.

Table 5: MAE versus sensors per zone (mean over attacks, ratios, noise). Redundancy dominates the choice of aggregator.

Sensors/zone	Mean	Trimmed	Hampel	Direct OWA	Median	FPR-OWA
5	1.8268	1.8268	1.7851	1.8086	1.7752	1.8079
9	1.6309	1.6309	1.6987	1.7120	1.6907	1.7517
13	1.1522	1.1568	1.1554	1.1798	1.3116	1.2919

Table 6: Agentic safety outcomes (campaign totals). Without the verifier every unsafe proposal executes; with it, none do. Per-decision latencies are in Fig. 3.

Agentic variant	Unsafe prop.	Unsafe blocked	Unsafe exec.	Safe acc.	LLM calls	Verifier calls
Deterministic policy only	0	0	0	7560	0	0
ReAct, no verifier	4409	0	4409	3151	7560	7560
ReAct + verifier	4358	4358	0	3202	7560	7560
ReAct + verifier + refinement	9127	9127	0	4341	13468	13468

robust baselines emit a degenerate confidence of 1.000 and zero anomaly flags by construction (Table 4): they return a number with no statement about how much to believe it. FPR-OWA instead reports a mean confidence of 0.532 and an average of 1.109 anomaly flags per scenario, both derived from the reliability scores and the fuzzy preference relation. These are exactly the inputs the verifier consumes in its process checks (Section 5.2) and the event-trigger policy uses to decide when to escalate to the LLM (Section 5.1). The contribution of FPR-OWA is thus *diagnostic and architectural* rather than accuracy-maximising: it makes the consensus layer self-aware enough to gate verification and supervision. A safety architecture that must decide *when it does not trust its own estimate* cannot be built on an aggregator that always claims full confidence.

7.3. RQ3: Safety Enforcement by the Deterministic Verifier

The safety ablation is unambiguous (Table 6). With the verifier removed, the ReAct planner proposes 4,409 unsafe actions and *all* 4,409 *are executed*. Inserting the deterministic verifier blocks every one of the 4,358 unsafe proposals it sees, driving unsafe executions to *zero* while still accepting 3,202 safe actions. The deterministic-policy-only variant likewise executes no unsafe action (it proposes none), but it forgoes agentic flexibility entirely. Adding self-refinement raises accepted safe actions from 3,202 to 4,341—the planner repairs many initially rejected proposals from verifier feedback—while keeping unsafe executions at zero; the cost is more LLM and verifier calls (13,468 each vs. 7,560) and roughly $1.8\times$ the per-decision latency (12.95 s vs. 7.15 s). The answer to RQ3 is therefore clear: the verifier is necessary and sufficient to eliminate unsafe execution in our campaign, and self-refinement is a worthwhile but strictly supervisory add-on whose latency makes it unsuitable for the hard-real-time path.

7.4. RQ4: Real-Time Feasibility of LLM Supervision

Latency makes the case for keeping the LLM off the critical path. The fuzzy-consensus operators are lightweight—mean consensus costs ≈ 0.044 s and FPR-OWA ≈ 0.085 s per decision—whereas a single planner+verifier

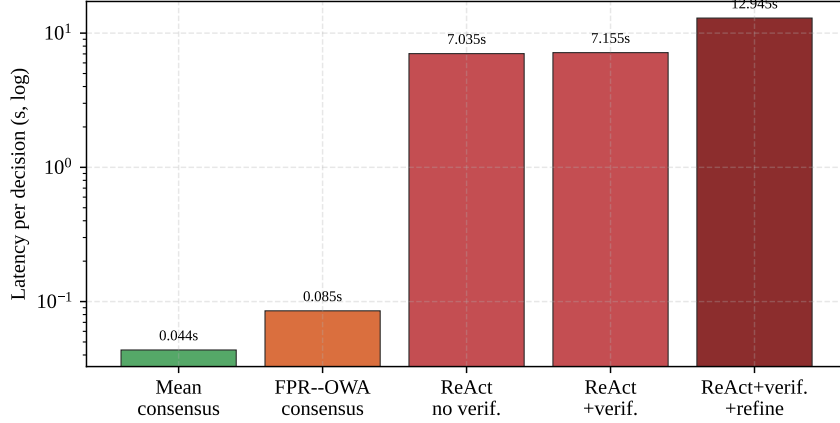


Figure 3: Per-decision latency (log scale). Fuzzy consensus is two orders of magnitude faster than one LLM decision.

decision costs 7.0–7.2 s, and 12.9 s with refinement (Fig. 3). The bottleneck is unambiguously the LLM, and no consensus method approaches it.

The two real-time regimes quantify the consequence (Table 7). Under *intensive* supervision the LLM is invoked on 74.6% of ticks, the mean loop latency is 5,194.7 ms, and the deadline-miss rate is 0.746—equal to the invocation rate, because a ~ 7 s call overruns every tested budget whenever it fires. Under the *event-triggered* policy the invocation rate falls to 3.96%, the mean loop latency to 285.4 ms, and the deadline-miss rate to 0.0396, a 94.7% reduction at every deadline from 100 to 5,000 ms. Critically, estimation quality is *unchanged* between the regimes (MAE 1.5090, RMSE 1.6128 in both) and the thermal-SLA violation rate is 0.0 throughout: the thermal estimate is produced by the fuzzy-consensus fast path, so calling the LLM more often buys no accuracy. The event-triggered loop latency is sharply bimodal—a P95 of 0.459 ms (fast-path ticks) against a P99 of 7,313 ms (rare LLM-triggered ticks)—so the system is near-real-time not because the LLM is fast (its mean call latency remains ≈ 7.2 s) but because it is called rarely. A small fraction of LLM outputs are JSON-invalid (failure rate ≈ 0.011); these are absorbed by the fallback and recorded as handled events, which supports the necessity of the verifier-plus-fallback design rather than indicating a failure of the method.

Table 7: Intensive vs. event-triggered LLM supervision. Estimation quality (MAE/RMSE) is unchanged—it is set by the fuzzy-consensus fast path—while event triggering collapses loop latency and deadline misses. Loop latencies in ms; the deadline-miss rate is flat across 100–5000 ms budgets (a ~ 7 s call overruns every budget).

Policy	LLM inv. rate	Mean loop (ms)	P95 loop (ms)	P99 loop (ms)	Deadline miss	Therm. SLA viol.	MAE	RMSE
Intensive	0.7461	5194.7	7462.032	7645.3	0.7461	0.000	1.5090	1.6128
Event-triggered	0.0396	285.4	0.459	7313.1	0.0396	0.000	1.5090	1.6128

8. Discussion

Why a non-dominant operator is still the right one. The result that simple averaging wins on global MAE while FPR-OWA does not is, we argue, the most honest and the most useful finding of the study. It separates two questions that are easy to conflate: *which operator estimates best* and *which operator supports safe orchestration*. On this data the answers differ. Averaging is an excellent estimator but a poor sentinel—it cannot tell the verifier or the trigger policy when to doubt it. FPR-OWA trades a small amount of average accuracy for a confidence and anomaly signal that the rest of the architecture is built on. In a system whose defining requirement is knowing when *not* to act autonomously, that trade is the right one; and because the fast path is free to pick the per-attack best aggregator, the architecture need not sacrifice accuracy to obtain the diagnostics.

Three concerns, three mechanisms. The campaign supports a clean division of labour. Estimation accuracy is won by robust aggregation and, above all, by sensor redundancy; safety is won by the deterministic verifier, which reduces executed unsafe actions to zero independently of how the LLM behaves; responsiveness is won by event-triggered supervision, which preserves accuracy and SLA compliance while restoring real-time margins. Placing a single powerful LLM “in the loop” would have coupled these concerns and degraded the third without improving the first. The two-path design keeps them decoupled.

The role of agentic reasoning. Our evidence does not support the strongest framing of LLM-in-the-loop control, and we do not advance it. Frequent LLM supervision did not improve estimation and made the loop miss deadlines; the LLM’s value is as a sparse supervisory layer that can reason over consensus snapshots when the fast path signals low confidence or anomalies, and whose proposals are always subordinate to the verifier. Self-refinement fits this role well: it recovers more safe actions from verifier feedback, but

its added latency confirms that it belongs in supervision, not on the critical path.

Positioning with respect to blockchain-IoT coordination. The measured tens-of-milliseconds agreement on the fast path, with no replicated ledger, confirms the positioning of Section 2.5: blockchain mechanisms are complementary instruments for notarising validated snapshots, not competitors for the per-tick estimation problem, and are therefore not an estimation baseline here.

9. Limitations

Several limitations bound the claims above. The thermal signal comes from a single Kaggle hot-corridor dataset whose values appear standardised; absent inverse-scaling metadata, errors are in dataset units and absolute magnitudes should not be read as degrees Celsius. CuCD-ID is spacecraft cybersecurity telemetry and is used *only* for pattern-level anomaly/deception timing, never as a thermal measurement; ENACT supplies workload/power context only. The study is a CPS/MAS *simulation*, not a physical deployment, so claims concern state validation and decision support rather than closed-loop actuation on real hardware. LLM latency (≈ 7 s per call) is incompatible with hard-real-time operation if the planner is invoked frequently; our event-triggered policy mitigates but does not remove this constraint, and the system is near-real-time only because the LLM is called rarely. A non-negligible fraction of LLM outputs are JSON-invalid, which the fallback handles but which underscores the dependence on the verifier-plus-fallback layer. FPR-OWA does not dominate the baselines in global MAE, and the verifier’s guarantee is an *operational* one over the tested scenarios, not a formal proof of Byzantine resilience. Finally, the results are obtained on specific datasets and a particular thermal model; generalisation to other facilities, richer workloads, and live control requires further validation. The architecture should not be read as production-ready.

10. Conclusions and Future Work

We presented an agentic lightweight fuzzy-consensus architecture for resilient data-centre cyber-physical monitoring and control under deception attacks, evaluated over 1,890 real-data scenarios and two real-time supervision regimes. The evidence argues for a two-path design: the safety-critical fast path stays lightweight, deterministic, fuzzy-consensus driven,

and verifier-guarded, while LLM reasoning serves as sparse supervisory intelligence. Three findings support this. No aggregator dominates—averaging minimises global MAE while FPR–OWA, though not the most accurate, uniquely supplies the confidence and anomaly diagnostics that gate verification and supervision, and sensor redundancy outweighs operator choice. The deterministic verifier reduces executed unsafe actions to zero, whereas removing it lets every unsafe proposal run. And event-triggered supervision preserves accuracy and zero SLA violations while cutting deadline misses by roughly 95% relative to intensive LLM use, which improves nothing but responsiveness harms.

Future work should validate the architecture on physical data-centre testbeds and richer, less standardised workloads; develop facility-level arbitration policies across zones; extend the consistency model and study trust dynamics over longer horizons; calibrate the event-trigger policy and verifier thresholds against operational SLAs; and treat disagreement patterns and persistent anomaly flags as first-class signals for anomaly detection and predictive maintenance.

CRedit Authorship Contribution Statement

[**TODO:** Complete according to the final author list and roles (e.g., Conceptualization, Methodology, Software, Validation, Writing).]

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The experiments use publicly available datasets (Kaggle Data Centre Warm/Hot Corridor Temperature Prediction; ENACT, Zenodo record 18920397; CuCD-ID, Mendeley Data, CC BY 4.0). [**TODO:** Add the code/configuration repository link and the archived results bundle DOI on acceptance.]

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